EX: NO: 07

SLIDING WINDOW PROTOCOL

AIM:

Write a program to implement flow control at data link layer using Sliding window Protocol.

PROGRAM:

```
import time
message = input ("Enter the text message: ")
window_size = int (input ("Enter window size: "))
error_frame = int (input ("Enter frame: "))
if (window_size <=0:
        print (" Invalid window size")
        exit()
print (" SLIDING WINDOW PROTOCOL")
print ("In Message :", message)
print ("window size", window_size)
start = 0
expected_frame = 1
error_induced = false
while start < len (message):
        end = min (start + window_size, len (message)
        print ("SENDER BUFFER")
        sender_buffer = []
        for i in range (start, end):
                frame_no = i+1
                data = message[i]
        sender_buffer.append ([frame_no, data])
        print ("frame_NO =", frame_no, "DATA =", data)
print ("In Sending frame :")
time.sleep (1)
print ("RECEIVER")
error_found = false
for frame in Sender_buffer:
        frame_no = frame [0]
```

```
        data = frame[i]
        if frame_no == error_frame and not error_induced:
            received_frame = frame_no + 1
            error_induced = True
        else:
            received_frame = frame_no
        print("Received frame:", received_frame,
              "DATA:", data)
        if received_frame == expected_frame:
            print("frame", received_frame, "is correct")
            expected_frame += 1
        else:
            print("In ERROR DETECTED")
            print("Expected from:", expected_frame)
            print("Received frame:", received_frame)
        print("RECEIVER BUFFER")
        print("NACK =", expected_frame)
        print("Resending frames from frame", expected_frame)
        error_found = True
        break
    if error_found:
        start = expected_frame - 1
        time = sleep(1)
        continue
print("RECEIVE BUFFER")
ack = expected_frame
print("Ack =", ack)
print("In Ack received by Sender")
print("Expected Ack:", end+1)
```

```
if Ack == end+1:
        print("\n Ack is correct"):
        print(" Sliding window moves forward")
        start = end.
else:
        print ("\n Incorrect Ack ")
        print ("Waiting for correct Ack")

Print (" Transmission Completed)
Print ("\n Original Message:", message)
Print ("All frames successfully transmitted")
```

OUTPUT:

```
Enter the text message : hellow
Enter window size :
                SLIDING WINDOW PROTOCOL
Message : hello
Window size : 3
                SENDER BUFFER
frame_NO = 1    DATA = h
frame_NO = 2    DATA = e
frame_NO = 3    DATA = l
Sending frame ...
                RECEIVER
Received frame : 1    DATA: h
Frame 1 is Correct
Received frame : 2    Data : e
Frame 2 is Correct
Recieved Frame : 8    Data : l
                Error Detected
Expected frame : 3
Received frame : 8
                TRANSMISSION COMPLETED
Original Message : hello
All frames Sucessfully transmitted
```

RESULT:

Thus flow control at data link layer using SLIDING WINDOW PROTOCOL